



COLLEGE OF ENGINEERING, PUNE

(An Autonomous Institute of Government of Maharashtra.)

Programme: B.Tech
Course Code: -
Branch: Computer Engineering
Duration: 40 mins
Student PRN No.

T2 Examination

Semester: IV
Course Name: Data Structures II
Academic Year: 2021 -2022
Max Marks: 10

1	1	2	0	0	3	0	3	1
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Instructions:

1. Write your PRN Number on Question Paper.
2. Draw neat and clear diagrams wherever required.
3. Write code in C - Programming Language

1.

ANSWER ANY ONE

- a. Write `insert()` functions for ADT heap implemented using Array
- b. Write `insert()` function for inserting a node in a Red-Black Tree.
- c. Write `delete()` function to delete an element from a min-heap.

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2.

ANSWER ANY TWO

- a. Write a recursive function `int find_Max(Tree *t)` to return maximum element in a Binary Tree.
- b. Write a recursive function `int count(Tree *t)` to count number of internal nodes in a Binary Tree.
- c. Write recursive function: `int is_Heap(Tree *t)` to verify the given Complete Binary Tree is a heap or not.

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each